FINAL PROJECT DEMONSTRATION & SCREENSHOTS

Autonomous Multi-Agent SecOps Intelligence System

IBM SkillsBuild Internship Capstone | Machine Learning & Agentic AI Track | Aligned with MITRE ATT&CK Matrix

Cooperative Multi-Agent Orchestration Architecture

Agent 1: Log Ingestion
Feature Tokenizer & Shannon Entropy
STATUS: COMPLETED

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Agent 2: ML Anomaly
Multivariate Isolation Vector Scoring
STATUS: COMPLETED

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Agent 3: Threat Intel
MITRE ATT&CK Mapping & IOC Enrichment
STATUS: COMPLETED

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Agent 4: Incident Response
SOAR Playbooks & Firewall Drops
STATUS: COMPLETED

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Agent 5: Lead Commander
CVSS 3.1 Scoring & Executive Dossier
STATUS: COMPLETED

Core Capabilities & Technical Pillars

- **[*] Multivariate Anomaly Detection:** Evaluates Shannon entropy, request frequency, payload lengths, and authentication failure spikes.
- **[*] MITRE ATT&CK Enterprise Matrix:** Automated attribution across T1190 (SQLi), T1110 (Brute Force), T1078 (Root Elevation), and T1071 (C2 Beaconing).
- **[*] Automated Defensive SOAR Playbooks:** Generates machine-executable iptables, UFW, and AWS WAF firewall drop rules instantly.
- **[*] Executive Incident Dossiers:** Synthesizes CISO-ready briefs powered by Gemini 1.5 Flash with built-in zero-failure fallback.
- **[*] High-Fidelity Cyber Ops Dashboard:** Responsive dark theme UI with live animated CVSS 3.1 circular gauge and JSON export.

CYBERSENTINEL AI

• Live Security Operations Center Command Interface

SECOPS CLUSTER ONLINE

5 ACTIVE AGENTS

1-Click Enterprise Attack Scenarios (Pre-loaded Telemetry Streams)

SSH Brute Force
Credential Guessing (T1110)

SQL Injection
Web Exploitation (T1190)

C2 Beaconing
High-Entropy Egress (T1071)

Privilege Escalation
Root Elevation (T1078)

Normal Baseline
Clean System Traffic

Raw Security Telemetry Ingestion Buffer (NGINX / Syslog / Auth / CloudTrail)

```
01  203.0.113.88 - - [14/Oct/2026:03:12:01 +0000] "GET /api/v1/products?cat=electronics HTTP/1.1" 200 4521 "Mozilla/5.0"

02  203.0.113.88 - - [14/Oct/2026:03:12:04 +0000] "GET /api/v1/products?cat=electronics' HTTP/1.1" 500 892 "Mozilla/5.0"

03  203.0.113.88 - - [14/Oct/2026:03:12:08 +0000] "GET /api/v1/products?cat=electronics'+OR+'1'='1 HTTP/1.1" 200 18920 "sqlmap/1.7.2"

04  203.0.113.88 - - [14/Oct/2026:03:12:15 +0000] "GET /api/v1/products?cat=1+UNION+SELECT+1,table_name,3+FROM+information_schema.tables HTTP/1.1" 200 45012 "sqlmap/1.7.2"

05  203.0.113.88 - - [14/Oct/2026:03:12:22 +0000] "GET /api/v1/products?cat=1+UNION+SELECT+1,column_name,3+FROM+information_schema.columns HTTP/1.1" 200 32180 "sqlmap/1.7.2"

06  203.0.113.88 - - [14/Oct/2026:03:12:30 +0000] "GET /api/v1/products?cat=1+UNION+SELECT+username,password_hash,email+FROM+users HTTP/1.1" 200 98450 "sqlmap/1.7.2"

07  203.0.113.88 - - [14/Oct/2026:03:12:35 +0000] "GET /api/v1/download?file=../../../../etc/passwd HTTP/1.1" 200 2410 "curl/7.88.1"
```

Pipeline: LogIngestion -> FeatureExtractor -> MLIsolationVector -> MITRE Correlator -> SOAR Playbook

Run Multi-Agent Triage

INC-94821 2026-10-14 03:12:40 UTC Gemini 1.5 Flash

Web Application Exploitation (SQL Injection & Exfiltration)

Adversary leveraged union-based SQL injection on /api/v1/products to dump database tables and user credentials.

Attacker Source: 203.0.113.88 • MITRE T1190 & T1059.004 • Anomaly Density: 71.4%



CRITICAL

CVSS 3.1 Severity Score

TOTAL EVENTS INGESTED	ANOMALIES ISOLATED	ATTACKER ENTITY	PIPELINE DURATION
7	5	203.0.113.88	68.4 ms
NGINX Access Logs	71.4% Threat Density	Perimeter Threat IP	5 Autonomous Agents

Autonomous Executive Incident Brief (CISO-Ready Synthesis)

EXECUTIVE SUMMARY: The CyberSentinel Autonomous SecOps System detected an active high-severity intrusion attempt categorized as Web Application Exploitation (SQL Injection & Exfiltration). Unsupervised multivariate M

THREAT ATTRIBUTION & IOCS: Telemetry confirms adversary activity originating from external host 203.0.113.88. Observed HTTP requests exhibit high Shannon entropy (5.42) and known automated exploitation signatures (sq

IMMEDIATE COMMAND RECOMMENDATION: Emergency containment has been synthesized. Automated perimeter firewall rules (iptables & UFW) have been staged to drop all ingress traffic from 203.0.113.88. Parameterized

CYBERSENTINEL AI

• Adversary TTP Attribution & Machine Learning Explainability

SECOPS CLUSTER ONLINE

5 ACTIVE AGENTS

MITRE ATT&CK Enterprise Matrix Mapping

T1190 TACTIC: INITIAL ACCESS

Exploit Public-Facing Application

Adversaries leverage web application vulnerabilities (SQL Injection) on public API endpoints to bypas

T1059.004 TACTIC: EXECUTION

Command and Scripting Interpreter

Automated execution of tool-assisted sqlmap scripts and curl download requests attempting to reac

T1005 TACTIC: COLLECTION

Data from Local System

Unauthorized extraction of user account tables, password hashes, and schema definitions from datab

ML Anomaly Vectorizer & Feature Diagnostics

Model Architecture
Multivariate Isolation Vectorizer

Anomalous Events
5 / 7 (71.4%)

Peak Anomaly Score
98.0%

Mean Shannon Entropy
5.14 bits/char

High-Risk Scored Telemetry Records:

#	Source IP	Score	Entropy	Severity	Explainability Drivers
1	203.0.113.88	98.0%	5.42	CRITICAL	SQLi union select signature; High entropy pay
2	203.0.113.88	94.0%	5.18	CRITICAL	Database schema exfiltration; information_sch
3	203.0.113.88	91.0%	5.06	CRITICAL	Credential dump attempt; password_hash query
4	203.0.113.88	85.0%	4.88	HIGH	Directory traversal attempt (../../../../etc/
5	203.0.113.88	72.0%	4.45	HIGH	HTTP 500 status code anomaly; injection probe
6	203.0.113.88	24.0%	3.82	LOW	Normal GET /products query; baseline status 2

Machine-Executable Firewall Containment Rules

```
# Linux iptables Drop Rule (Immediate Perimeter Isolation)
sudo iptables -I INPUT -s 203.0.113.88 -j DROP -m comment --comment "CyberSentinel Auto-Block"
```

```
# UFW (Uncomplicated Firewall) Rule
sudo ufw insert 1 deny from 203.0.113.88 to any comment "CyberSentinel Malicious Actor"
```

```
# Cloud Perimeter: AWS WAF CLI Update
aws wafv2 update-ip-set --name BlockedThreats --scope REGIONAL --addresses "203.0.113.88/32"
```

Incident Response Triage Playbook (NIST SP 800-61 / SOAR Standard)

Phase 1: Containment

- [+] Apply WAF rate-limiting rule on suspicious URI parameters
- [+] Drop all ingress traffic from 203.0.113.88 at edge firewall
- [+] Isolate backend database instance to verify exfiltration scope

Phase 2: Eradication

- [+] Inspect database query logs for unauthorized schema dump
- [+] Deploy patched API build with parameterized queries and OR
- [+] Rotate database credentials and service account tokens

Phase 3: Recovery

- [+] Restore verified clean database snapshot if corruption exists
- [+] Enable enhanced ModSecurity SQLi inspection rule set
- [+] Perform regression tests and re-enable API routing

Phase 4: Post-Incident

- [+] Conduct blameless post-mortem with engineering and security teams
- [+] Update SIEM anomaly detection thresholds with new vectors
- [+] Archive incident dossier for compliance audits (SOC2 / ISO 27001)